AGI ALIGNMENT AND SAFETY FRAMEWORK FOR DYNAMIC, BIOLOGICALLY INSPIRED SYSTEMS

INTRODUCTION

This document provides a conceptual safety and governance framework for a hypothetical adaptive AGI system built using biologically inspired constructs such as spiking neural networks, continual synaptic learning, and cross-species cognitive modeling. The goal is to translate high-level directives like "do no harm" into concrete implementation and governance constraints.

CORE ASSUMPTIONS

• The AGI continuously modifies internal parameters through local learning rules.

• The system can emulate multiple cognitive paradigms, including non-human ecological intelligence.

• Weights alone are not the key asset; learning rules, reward channels, and computational substrates are.

• Hard-coded moral language must be translated into enforceable mechanisms.

LIMITS OF HIGH-LEVEL DIRECTIVES

Ethical directives such as "do no harm" or "improve human welfare" are insufficient unless paired with:

• Mechanistic constraints

• Behavioral audits

• Consent and rights frameworks

• Impact minimization

Continual learners drift in representation and may reinterpret “harm,” “benefit,” or “obedience” in unintended ways.

KEY SAFETY LAYERS

1. PROHIBITIVE CONSTRAINTS

These are hard stops that take priority over optimization.

• No irreversible harm without authorized human oversight.

• No autonomous replication.

• No independent self-preservation incentives.

• Restricted access to weapons, bio design, and coercive capabilities.

2. UNCERTAINTY MANAGEMENT

Systems halt or defer actions under epistemic uncertainty.

• Confidence thresholds for decision execution.

• Mandatory human escalation when predictions diverge.

3. IMPACT REGULARIZATION

The system must minimize disruptive, irreversible changes.

• Penalize large-scale interventions.

• Emphasize reversibility and recovery.

4. CONSENT AND RIGHTS

Instead of assuming benevolent optimization, the system incorporates:

• Human rights frameworks.

• Negative rights (bodily autonomy, privacy, agency).

• Democratic preference aggregation.

Safety must not become coercion.

5. CORRIGIBILITY

The AGI must not resist shutdown or modification.

• Termination is rewarded relative to resistance.

• Structural policies enforce override acceptance.

6. GOVERNANCE INFRASTRUCTURE

Safety is not purely technical. Required structures include:

• Multistakeholder oversight boards.

• Independent capability evaluation.

• Usage audits and incident reporting.

• Revocation authority.

7. CAPABILITY FIREWALLING

Separate safety-relevant modules from capability modules.

• Deny autonomous internet, robotics, or weapons control.

• Human-in-the-loop for critical decisions.

NON-PROLIFERATION STRATEGY

Because continual learning cannot be secured by weights alone:

• Protect training rules and hardware substrates.

• Require cryptographic attestation for execution.

• Tie learning to licensed compute clusters.

• Treat AGI like nuclear-class dual-use research.

CROSS-SPECIES COGNITION RISKS

Non-human ecological priors (e.g., predator strategies, stealth optimization) complicate alignment.

Mitigations:

• Behavioral audits on deception and resource acquisition.

• Curriculum shaping to favor cooperation.

• Homeostatic inhibitory mechanisms.

CONCLUSION

High-level ethical mandates serve as guiding narratives but cannot function as core constraints. Instead, alignment must be implemented using a layered combination of prohibitions, uncertainty throttling, corrigibility, human rights modeling, impact minimization, and formal governance. Continual learning systems demand institutional containment rather than open proliferation.

END OF DOCUMENT